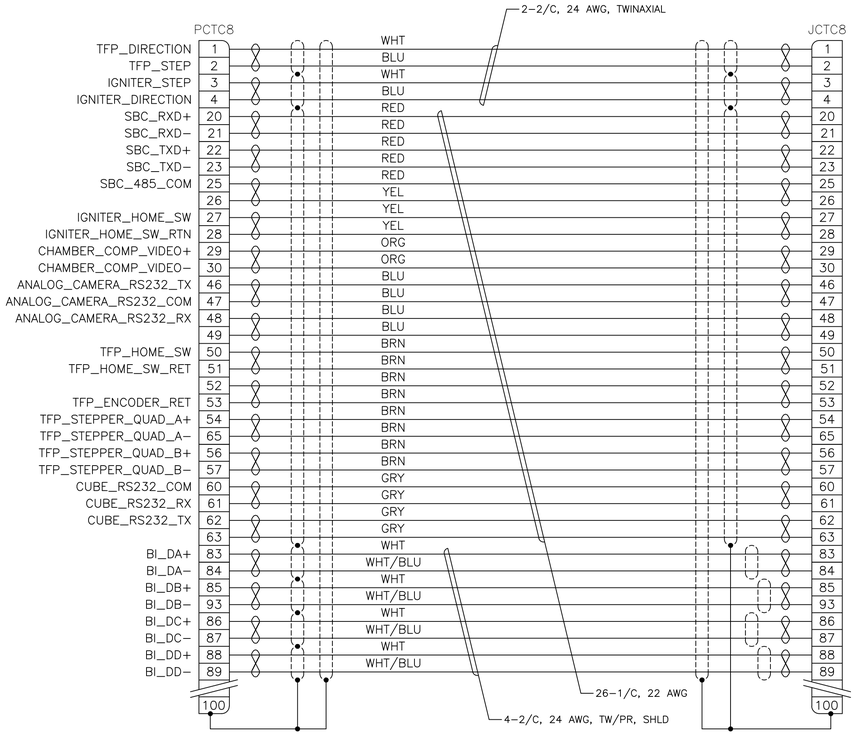
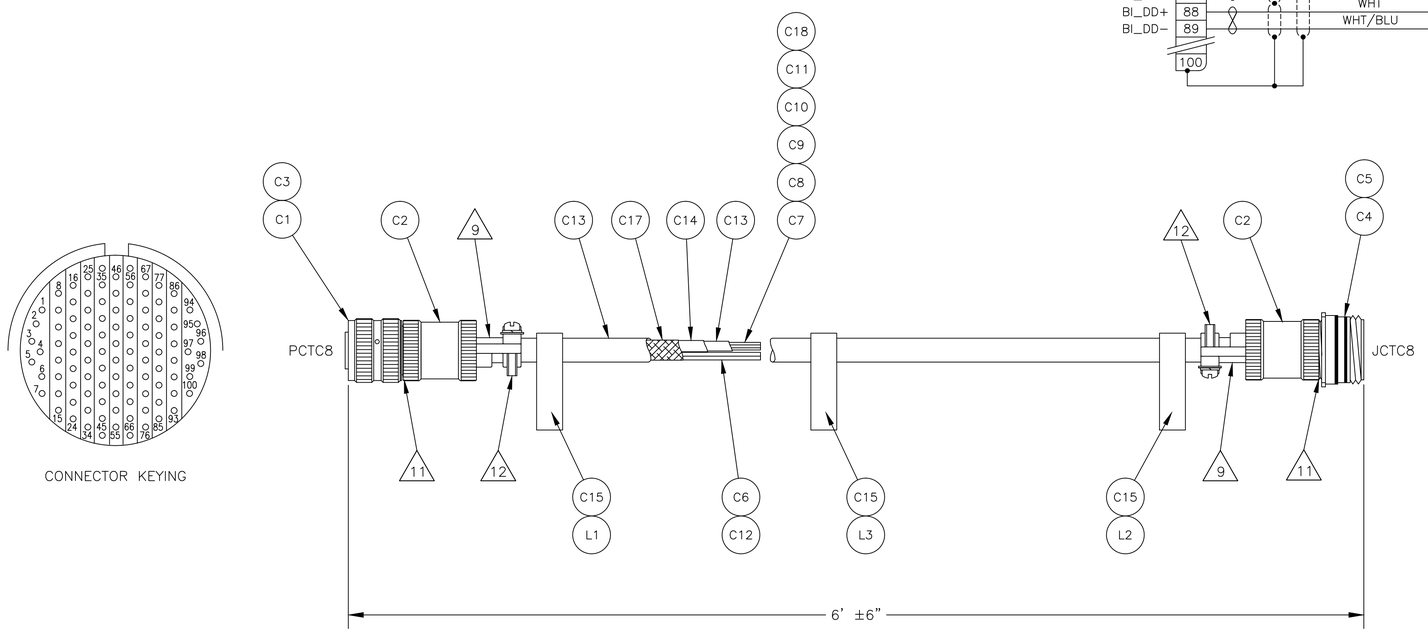
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	1	D38999/26FH35PC	CONNECTOR, CIRCULAR PLUG 100 POS	QPL
C2	2	380HS115UC2312A3	BACKSHELL, STRAIGHT	GLENAIR
C3	A/R	M39029/58-360	CONTACT, PIN SIZE 22D, AWG 22-28	QPL
C4	1	D38999/20FH35SC	CONNECTOR, CIRCULAR RECEPT., 100 POS	QPL
C5	A/R	M39029/56-348	CONTACT, SOCKET SIZE 22D, AWG 22-28	QPL
C6	A/R	M27500-24RC2S06	2/C, 24 AWG, TWISTED, SHIELDED, JACKETED	QPL
C7	A/R	M22759/11-22-1	1/C, 22 AWG, PTFE INSULATED, BROWN	QPL
C8	A/R	M22759/11-22-2	1/C, 22 AWG, PTFE INSULATED, RED	QPL
C9	A/R	M22759/11-22-4	1/C, 22 AWG, PTFE INSULATED, YELLOW	QPL
C10	A/R	M22759/11-22-6	1/C, 22 AWG, PTFE INSULATED, BLUE	QPL
C11	A/R	M22759/11-22-8	1/C, 22 AWG, PTFE INSULATED, GRAY	QPL
C12	A/R	M17/176-00002	2/C, 24 AWG, TWINAXIAL	QPL
C13	A/R	5011001332S	EXPANDABLE SLEEVING 1/2" BLACK	BENTLEY HARRIS
C14	A/R	2172	BRAID, EMI SHIELDING, 3/8"	ALPHA
C15	A/R	TZE-211	LABEL, P-TOUCH, 1/4", BLACK ON WHITE	BROTHER
C16	A/R	P224X.5	TAPE, KAPTON, 1/2" WIDE	PERMACEL
C17	A/R	2146	BRAID, EMI SHIELDING, 1/2"	ALPHA
C18	A/R	M22759/11-22-3	1/C, 22 AWG, PTFE INSULATED, ORANGE	QPL

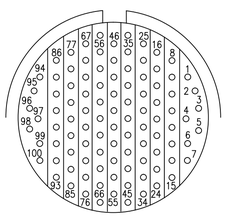
LABEL LIST	
ITEM	INSCRIPTION
L1	PCTC8
L2	JCTC8
L3	S1101EGA4562 SN XXXX



SCHEMATIC



CONNECTOR KEYING



CONNECTOR KEYING

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.3.
4. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
5. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND THE FOLLOWING NOTES:
   a. ALL WIRES 22 AWG UNLESS SPECIFIED.
   b. TO PREVENT POTENTIAL DAMAGE (COLD FLOW) OF THE UNDERLYING WIRE INSULATION, SEPARATOR SHALL BE APPLIED OVER THE WIRE BUNDLE TO GIVE THE WIRE CONTINUOUS PROTECTION.
   c. ALL SHIELD TERMINATIONS ARE INTERFERENCE SENSITIVE.
   d. WIRE TWIST SHALL BE BETWEEN 8 AND 16 TIMES THE OUTER DIAMETER OF THE HARNESS UNLESS OTHERWISE NOTED. (FROM NASA-STD-8739.4 PARA 7.3.20)
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTORS.
7. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
8. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.
9. USE KAPTON TAPE (ITEM C16) TO SECURE EXPANDO SLEEVE (ITEM C13) WITHIN BACKSHELL.
10. TERMINATE SHIELDS INTO BACKSHELL PER MANUFACTURER'S SPECIFICATION.
11. TORQUE BACKSHELL (ITEM C2) TO 80 ±5 IN-LBS.
12. TORQUE FASTENERS PER CHART.
13. TEST PER NASA-STD-8739.4 USING THE FOLLOWING LEVELS:

    CONTINUITY: < 2.5 OHMS

    DWV: VOLTAGE: 1500VDC ±75VDC
    CURRENT: < 1 MILLIAMP
    DURATION: < 1 MINUTE

    IR: VOLTAGE: 500VDC ±50VDC
    RESISTANCE: > 100 MEG OHMS
    DURATION: < 1 MINUTE
14. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

CRIMP TOOL LIST				
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL	REF. DES.
M39029/58-360	M22520/2-01	M22520/2-09	M81969/14-04	PCTC8
M39029/56-348	M22520/2-01	M22520/2-07	M81969/14-01	JCTC8

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
QPL	M39029/58-360	22	13
		24	8
QPL	M39029/56-348	22	13
		24	8

BACKSHELL FASTENER TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 9/29/15
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 9/10/14	QUALITY	D. SEELEY 9/15/14
REVIEWER	J. BRUEWER 9/11/14	MATERIALS	M. WILKINSON 9/10/14
MECHANICAL	A. DRAKE 9/10/14	PROJ LEAD	B. BOROWSKI 9/15/14
ELECTRICAL	T. GOBELI 9/11/14		
STRUCTURAL	C. WERNER 9/10/14		
SAFETY	W. BORRELLI 9/10/14		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC14CA02C
PROJECT CODE: S2-BT3

REV	DESCRIPTION	APPROVED
A	REVISED PER ACME-ECO-382.	MH 9/28/15 DS 9/28/15 TG 9/29/15

REVISION HISTORY

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
PCTC8 EXTENSION CABLE
(PIL SIDE SIMULATION)

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EGA4562	A

SCALE	NONE	INITIAL REL DATE	09/16/14	SHEET	1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EFC3562_REV_A.DWG

AutoCAD 2012

REV A
SH 1
DWG NO S1101EGA4562